

Interview Q&A: scikit-learn

1. What is the main difference between K-Means and Hierarchical clustering?

K-Means is an unsupervised algorithm that partitions data into K clusters based on their similarity, whereas Hierarchical clustering builds a hierarchy of clusters by merging or splitting existing ones.

2. What is the purpose of StandardScaler in scikit-learn?

StandardScaler is used to scale numeric data to have a mean of 0 and a standard deviation of 1, which can improve the performance of some machine learning algorithms.

3. What is the purpose of cross-validation in scikit-learn?

Cross-validation is used to evaluate a model's performance on unseen data by splitting the available data into training and testing sets.

4. What is the difference between fit() and fit_transform() methods in scikit-learn's StandardScaler?

The fit() method computes the mean and standard deviation of the data, while fit_transform() also scales the data by subtracting the mean and dividing by the standard deviation.

5. What is the purpose of GridSearchCV in scikit-learn?

GridSearchCV is used to perform hyperparameter tuning for machine learning models by trying out different combinations of parameters and selecting the best one based on a scoring metric.

6. What is the purpose of the Pipeline class in scikit-learn?

The Pipeline class allows you to chain multiple estimators together, making it easier to manage complex workflows.

7. What is the purpose of the KNeighborsClassifier in scikit-learn?

The KNeighborsClassifier is a classifier that uses the k-nearest neighbors algorithm to predict the class label of new data points.

8. What is the difference between supervised and unsupervised learning in scikit-learn?

Supervised learning involves predicting a target variable based on input features, whereas unsupervised learning involves discovering hidden patterns or relationships in the data without a target variable.

9. What is the difference between a Decision Tree Classifier and a Random Forest Classifier in scikit-learn?

A Decision Tree Classifier uses a single decision tree to make predictions, while a Random Forest Classifier combines multiple decision trees to improve accuracy and reduce overfitting.

10. What is the purpose of the LabelEncoder in scikit-learn?

The LabelEncoder is used to convert categorical labels into numerical labels.

11. What is the purpose of the train_test_split function in scikit-learn?

The train_test_split function splits a dataset into training and testing sets, allowing for model evaluation and hyperparameter tuning.

12. What is the purpose of the OneVsRestClassifier in scikit-learn?

The OneVsRestClassifier is used for multi-class classification problems where each sample belongs to one class only.

13. What is the purpose of the SelectFromModel class in scikit-learn?

The SelectFromModel class is used to select features that are most relevant for a given model, based on their importance scores.

14. What is the purpose of the KFold class in scikit-learn?

The KFold class is used for splitting data into training and validation sets, allowing for k-fold cross-validation.

15. What is the difference between K-Means++ and Randomized K-Means initialization?

K-Means++ initializes centroids by selecting points with a probability proportional to their distance from the closest centroid, while Randomized K-Means randomly selects initial centroids.

16. What is the purpose of the metrics module in scikit-learn?

The metrics module in scikit-learn provides various evaluation metrics for classification, regression, clustering, and other tasks.

17. What is the purpose of the PCA class in scikit-learn?

The PCA class in scikit-learn is used for dimensionality reduction by retaining only the most informative principal components.

18. What is the purpose of the `make_classification` function in `scikit-learn`?

The `make_classification` function generates a random classification problem for testing purposes.